

OSINT Lab on TryHackMe

OSINT Dojo - Sakura Room

Background

This room is designed to test a wide variety of different OSINT techniques. With a bit of research, most beginner OSINT practitioners should be able to complete these challenges. This room will take you through a sample OSINT investigation in which you will be asked to identify a number of identifiers and other pieces of information in order to help catch a cybercriminal. Each section will include some pretext to help guide you in the right direction, as well as one or more questions that need to be answered in order to continue on with the investigation. Although all of the flags are staged, this room was created using working knowledge from having led and assisted in OSINT investigations both in the public and private sector.

Task 2: TIP-OFF

Objective: Find the attacker's username.

Methodology:

Methodology:

The task provides a link to an image file. The primary method to solve this is to analyze the image for hidden metadata which might contain information about its origin or creator.

1. Use an online metadata analysis tool, often referred to as an EXIF viewer. These tools are designed to extract hidden data embedded within files.

ID	svg8
Version	0.92.5 (2060ec1f9f, 2020-04-08)
Docname	pwnedletter.svg
Export-filename	/home/SakuraSnowAngelAiko/Desktop/pwnedletter.png
Export-xdpi	96
Export-ydpi	96
Metadata ID	metadata5
Work Format	image/svg+xml
Work Type	http://purl.org/dc/dcmitype/StillImage
Work Title	

So we found the username as SakuraSnowAngelAiko

Answer:

SakuraSnowAngelAiko

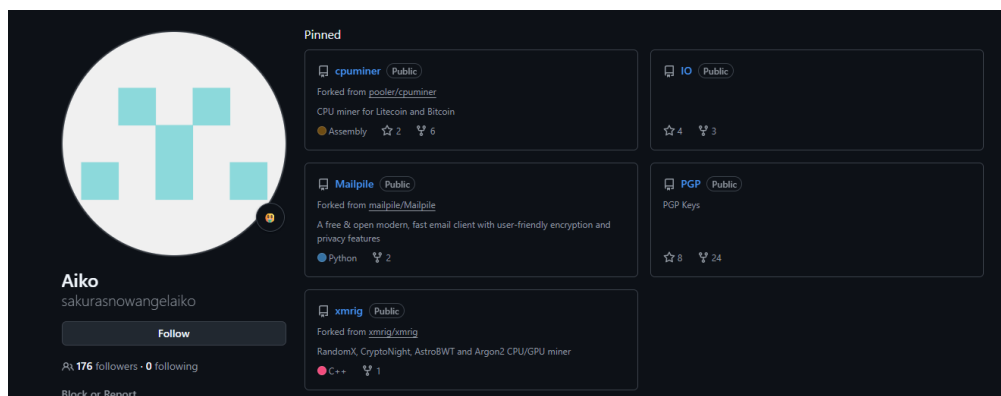
Task 3: RECONNAISSANCE

Objective: Find the attacker's email address and their full real name.

Methodology:

With a username, the next logical step is to use it as a search query across the internet to discover associated accounts.

1. A Google search for "SakuraSnowAngelAiko" reveals several key results, most notably profiles on **GitHub** and **LinkedIn**.
2. Navigating to the GitHub profile, we find a repository named PGP. PGP (Pretty Good Privacy) keys are used for encryption and often contain identifying information like a name and email address.
3. Inside the repository is a public PGP key. This key can be analyzed to extract the user ID associated with it. Using an online PGP decoder (like keybase.io's or others), we can paste the key's content.
4. The decoded key reveals the User ID: Aiko Abe <SakuraSnowAngel83@protonmail.com>.
5. This information is corroborated by the LinkedIn profile found earlier, which confirms the full name.



```

-----BEGIN PGP PUBLIC KEY BLOCK-----

mQNBGALrAYBDACSGmhCjKReLSBCNXmWVP5mN7saMKSkZDwGOCBBMVION52nqRyD
HiVLSmdwN2UwRXLF3oxCH5+QLxRpzrJlIkIgAXGD23z8ot+S7R7tZ8Yq2HvSe5J3L
FzoZjCph1VsvMfNIPYFcuFbwjJzvBAG00Js0rBj5t1EHaXK6rtJz6UMZ4n+82Vm9
Lix8VihIU9QfjGAYyVX735Z5ZMhEyNGQmusrDpahvIwjqEChVa4hyVIA0g7p5Fm
t6TzxhSPhNIPatCDIYL1wdonRDgQ3VrtG5S/dTNbzDgdvAg1388EEH00d+Vq0Tpu
fNr4GnKFep52czHVk8krWY1tLSzyVxHUFaSFYmH9FI2RU6Q5bCiHAIZKSP26mFeH
HPFmxrvStovco1s4f1t0A6bf+gbkkDj+MUGvrvUZwbeXbRvyOKTJNonhcf5bMz/D5
GSt0Ryd150+iiLLRyi5XF612RRHPfp7A4TsuH4+a0xoVaMxgCFZb7CMXNqDpeJ01
/idzm0HUKC1P6Z0AEQEAABoGU2FrdXJHU25vd0FuZ2VsODNACHJvdG9ubWpCbC5j
b22JAdQEEwEKAD4WIQSmUJZ8n0/iOkSaw9MXS3Q/SlBEEUACyAUSBgIBAwJA8Hp
uGULCQgHAGYVCgkICWIEFGIDAQIEAQIXgAAKCRD53Q/SlBEEUP/9C/0b6amQhTr7
0Jgf68KnS8nTXLJeoIS59+mOP/GVvw1dsfLoHkJYXuIc/fne2Y1y4qjVEDsCtAIS
rqReXnolyyqCNS2e70YsQ9Sgg0JG4o7r0VojKJNzuHDMQ944yhGk6zjC54qHba6+
37F9erDy+xRQ59B5gEFf2C60Fe00i+vpOWipqYAc1V6aUxHNRvYn8Fu01SIRTIo7
10LR1buhVgZvDIRR11dyFbF8B7oxrZZe9eWQURjXEVg07nh1V5UzekRv7qLsVyg
STV3mxodvXgw3KmrXU9FsFSKY9Cdu8vN9IVFJWQJj++rnzzyTUCUmxS89Y/L9wRX
4+7D5pfV1e4bG0ZKY+KQq1pYypUxIAFMheb2RKVjK5DzMDq6CQ573jqq/v1Ydp4
kNsucdZKEKn2evjJION750VEScus010jZuR93+w5Cmf4q60hpXSUT1AP016R1eue
8mPTmCra9dEmzAMsnLEPSPXNStzdxcdQhVvIDtj8M312iRyD6v1NeZa5AY0EYAU
BgEMAN4mk70jRDxwnjQd8AJ5133VncYT43gehVmkXaZ0AFaxoZtmR6oJbitWj+b1
fv1ILXP5LI803BZ2YPEVL8huqeFQjEIG4Suk3p/HuaIXavhiIJFRzoxoIZGM1Mh
XXRSq3Zd3LLg1Gir7smKSNv8qIlgNZr0TcphX9Qh90d/WqtCRYg5Rt8FibtkFI

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User ID Packet (0xd)

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cipherTypeByte: "180"
length: "32"
userId: "SakuraSnowAngel83@protonmail.com"

```

Signature Packet (0x2)

Answers:

- **Full Real Name:** Aiko Abe
- **Email Address:** SakuraSnowAngel83@protonmail.com

Task 4: UNVEIL

Objective: Find the attacker's cryptocurrency wallet address, their choice of crypto, and the pool that paid them on January 23, 2021.

Methodology:

The task description hints that the attacker has been deleting information from their GitHub, pointing us toward the commit history.

1. On the attacker's GitHub profile, we notice a repository named ETH, suggesting a connection to Ethereum.
2. Inspecting the ETH repository's commit history reveals a commit with the message "removed." This commit shows a deleted line of code.
3. The deleted line is a stratum connection string for a crypto miner:
stratum://0xa102397dbeeBeFD8cD2F73A89122fCdB53abB6ef.Aiko:pswd@eu1.ethermine.org:4444
4. This string contains the wallet address (0xa102...B6ef) and the mining pool (ethermine.org).
5. Using a blockchain explorer like **Etherscan.io**, we can input the wallet address to view its entire transaction history.

- The transaction history confirms that the wallet is primarily used for **Ethereum**, with some transactions involving **Tether (USDT)**.
- By filtering the transactions to the date **January 23, 2021**, we can see an "IN" transaction coming from the **Ethermine** mining pool.

```

1 - stratum://0xa102397dbeeBeFD8cD2F73A89122fCdB53abB6ef.Aiko:pswd@eu1.ethermine.org:4444
1 + stratum://ethwallet.workerid:password@miningpool:port

```

Comments 11

0x431503d5c7...	Transfer	11732875	1689 days ago	Ethermine	IN	0xa102397d...B53abB6ef	0.050069945 ETH	0.000021
0xde6bf2f4ee8...	Transfer	11714008	1692 days ago	Ethermine	IN	0xa102397d...B53abB6ef	0.050073325 ETH	0.000021
0x6559fcc264a...	Transfer	11686873	1696 days ago	0xa102397d...B53abB6ef	OUT	Tether: USDT Stabl...	0 ETH	0.0026209
0x480e838314...	Transfer	11684734	1696 days ago	0x35558d81...Da4c3F938	IN	0xa102397d...B53abB6ef	0.04805189 ETH	0.002037
0xc057104cc5...	Transfer	11678837	1697 days ago	0xa102397d...B53abB6ef	OUT	Tether: USDT Stabl...	0 ETH	0.00131045
0xf6fdb0b2d86...	Transfer	9806922	1986 days ago	0xa102397d...B53abB6ef	OUT	Tether: USDT Stabl...	0 ETH	0.00013111
0x5b3b8e1a48...	Transfer	9806917	1986 days ago	0xB256caa2...E2d6d2344	IN	0xa102397d...B53abB6ef	0.00045 ETH	0.000168

Show: 50 Records First < Page 1 of 1 > Last

Answers:

- Wallet Address:** 0xa102397dbeeBeFD8cD2F73A89122fCdB53abB6ef
- Choice of Crypto:** Ethereum and Tether
- Payment Pool (Jan 23, 2021):** Ethermine

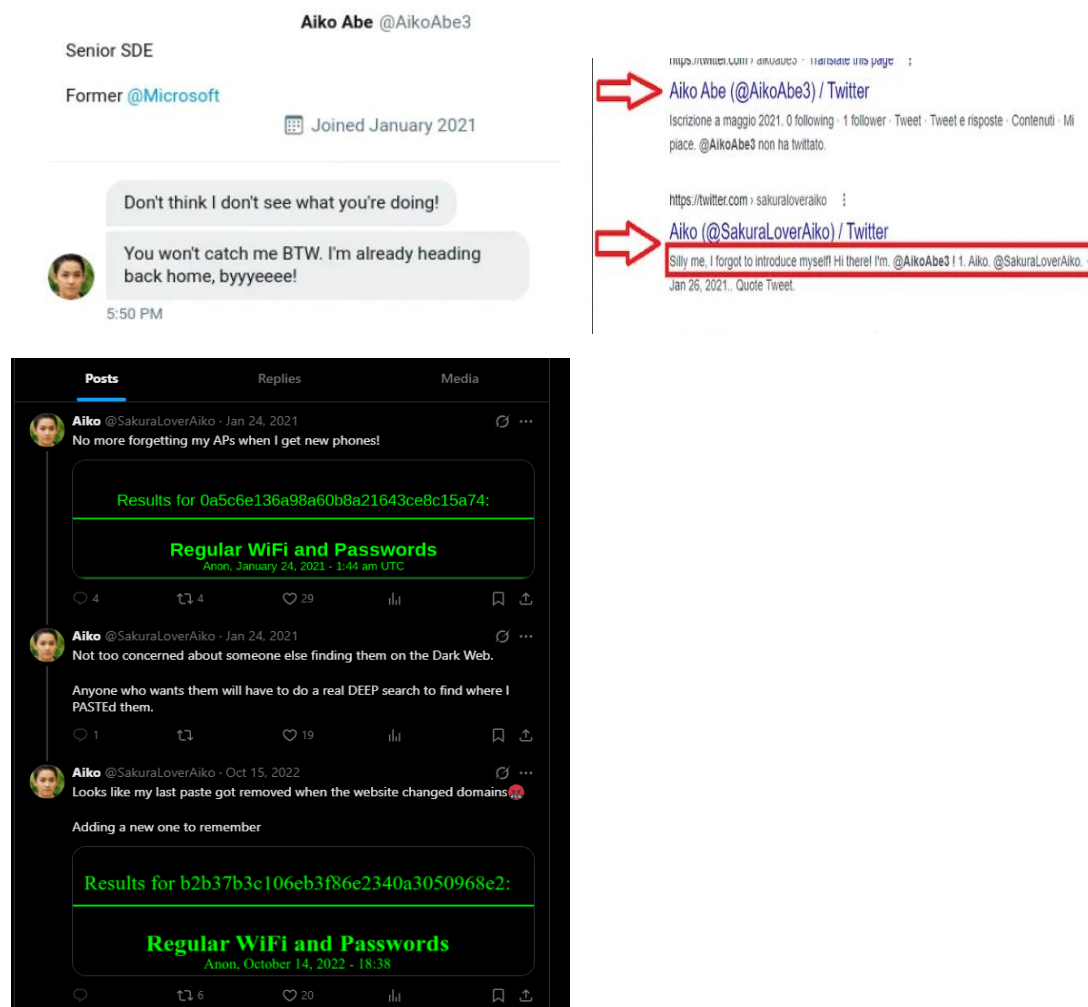
Task 5: TAUNT

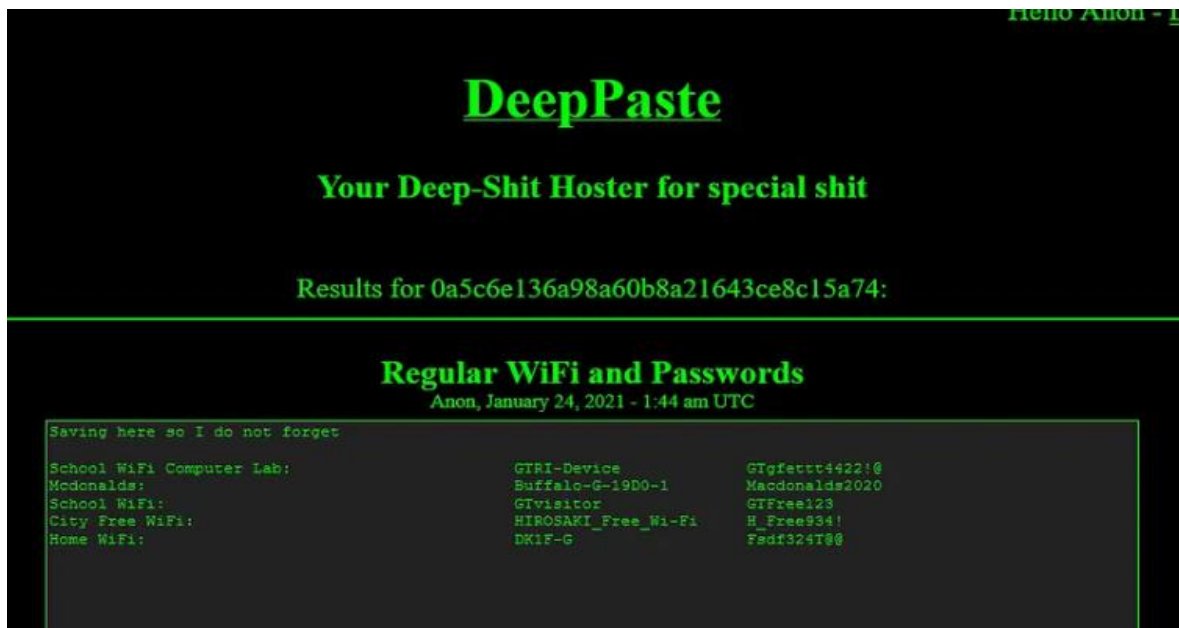
Objective: Find the attacker's current Twitter handle, the URL for their saved WiFi passwords, and the BSSID for their home WiFi.

Methodology:

This task begins with an image of a tweet where the original handle is crossed out.

1. The tweet in the image reveals the attacker's new Twitter handle.
2. Visiting this new Twitter profile, we find a post containing a screenshot of a paste on a dark web site called **DeepPaste**. The URL is partially visible.
3. Using the **Tor Browser** is necessary to access .onion websites. We navigate to the DeepPaste site and find the specific paste mentioned in the tweet (identifiable by its unique URL).
4. The paste contains a list of saved WiFi networks, including their SSIDs (network names) and passwords.
5. To find the BSSID (the MAC address of the router), we can use a wireless network mapping service like **Wigle.net**. The question asks for the BSSID of the "Home" network, which is listed in the paste with the SSID DK1F-G.
6. Searching for the SSID DK1F-G on Wigle.net provides a unique match, revealing its BSSID.





View Uploads Info Stats Tools omerwazap Log out

Network Search

General Search WiFi/Cell Detail Bluetooth Search

Average Location - Address

Num: 141 Street: West Jackson Boulevard

City: Chicago Region: IL

Country: US Postal: 60604

Average Location - Coordinates

Lat: 47.25264 to: 47.25265

Lon: -87.256243 to: -87.256244

Search Radius Tolerance(+/- degrees): 0.010

Network Characteristics

Last Observed: 20010925174546 Minimum data quality: 0 Encryption status:

BSSID/MAC: 0A2C:EF:3D:25:1B or 1st 3 Octets: 0A2C

SSID / Network Name (exact match): DK1F-G

SSID / Network Name (wildcards: % and _): foobar%

☐ Must Be a FreeNet ☐ Must Be a Commercial Pay Net ☐ Only Networks I Was the First to Discover

Query Reset

0-7 Product of number of observers and observations.
1 % means zero-or-more characters, _ means a single character.

showing records 1 to 1 of 1

Map	Net ID	SSID	Type	First Seen	Most Recently	Crypto	Est. Lat	Est. Long	Channel	Bcn Int.	QoS	Found by Me	Access	Comment
map	84:AF:EC:34:FC:F8	DK1F-G	infra	2019-04-24T11:00:00.000Z	2019-05-02T05:00:00.000Z		40.60551453	140.4606781	11	0	0	-	-	add comment

[more results](#)

Answers:

- **Current Twitter Handle:** SakuraLoverAiko
- **URL for WiFi SSIDs:**
depasteon6cqgrykzrgya52xglloh5ovyuyhte3ll7hzix7h5ldfqsyd.onion/show.php?md5=0a5c6e136a98a60b8a21643ce8c15a74
- **BSSID:** 84:AF:EC:34:FC:F8

Task 6: HOMEBOUND

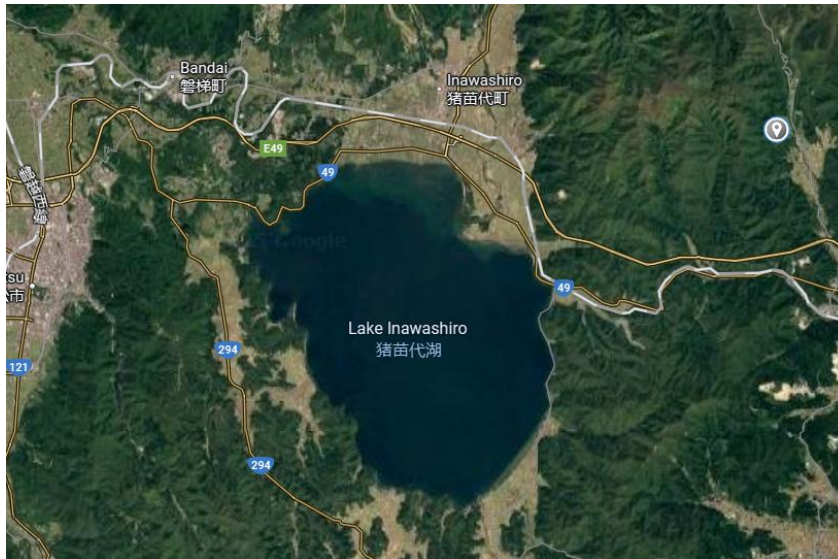
Objective: Find the airport closest to a posted picture, the layover airport, a lake seen on a map, and the attacker's home city.

Methodology:

This final task requires geolocating several images posted on the attacker's Twitter account.

1. **Closest Airport:** A photo on Twitter clearly shows the **Washington Monument** in Washington D.C., USA. A quick map search confirms the closest major airport is **Ronald Reagan Washington National Airport (DCA)**.
2. **Layover Airport:** Another image shows the interior of a first-class airport lounge. Using a reverse image search (Google Lens or Yandex), the image is identified as the **Japan Airlines Sakura Lounge** located at **Haneda Airport (HND)** in Tokyo.
3. **Lake Name:** A third image shows a flight path over a distinctively shaped lake. By comparing this shape to maps of Japan (the likely destination from HND), the lake is identified as **Lake Inawashiro**.
4. **Home City:** The home city can be deduced from the list of WiFi networks found in Task 5. One of the SSIDs is HIROSAKI_Free_Wi-Fi. This strongly implies the attacker considers **Hirosaki**, Japan, to be their home.





```
Saving here so I do not forget

School WiFi Computer Lab:      GTRI-Device      GTgfettt4422!@
Mcdonalds:                     Buffalo-G-19D0-1  Macdonalds2020
School WiFi:                    GTvisitor        GTFree123
City Free WiFi:                 HIROSAKI_Free_Wi-Fi  H_Free934!
Home WiFi:                      DK1F-G           FsdF324T@@
```

Answers:

- **Closest Airport:** DCA
- **Layover Airport:** HND
- **Lake Name:** Lake Inawashiro
- **Attacker's Home City:** Hirosaki